

# "Two firms will keep prices high." True?

A student argues that with only two sellers, both will simply hold price above cost and share the profit.

Is this right in the one-shot homogeneous model? If not, what must change?

► Reveal solution

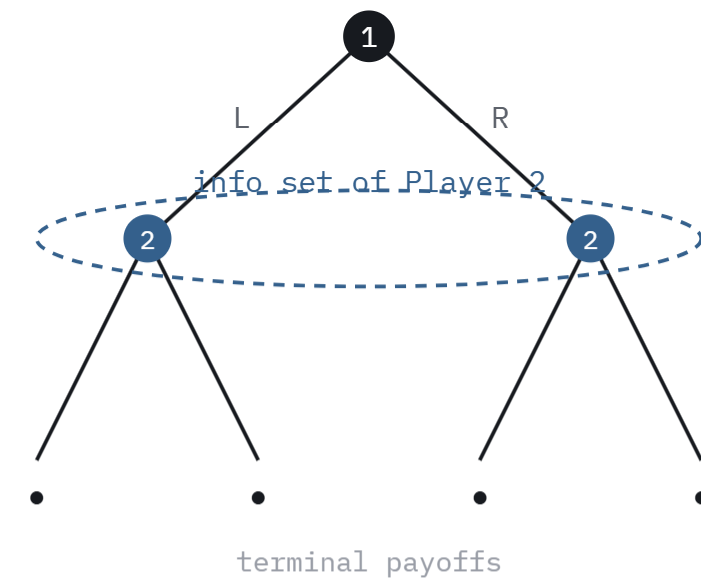
PART C

# Sequential & Finitely Repeated Games

Trees, backward induction, and subgame perfection — and why a known last period unravels cooperation all the way back to the start.

# Game trees & information sets

- ▶ Nodes (decision and terminal), branches (actions), and terminal payoffs.
- ▶ An information set groups nodes a player can't distinguish — consistency: the same action set at each node in the set.
- ▶ Perfect vs imperfect information: all singletons, or not.
- ▶ Perfect recall: a player never forgets their own past information or moves.



# Actions vs strategies, and Kuhn's theorem

- ▶ A strategy prescribes an action at every information set — including off-path nodes.
- ▶ So the pure-strategy count is the product over all of  $i$ 's information sets.
- ▶ Kuhn: with perfect recall, mixed and behavioral strategies are realization-equivalent — same distribution over terminal nodes.

PURE-STRATEGY COUNT

$$|S_i| = \prod_{h \in H_i} |A(h)|$$

Realization-equivalent, not loosely "payoff-equivalent": both induce the same distribution over outcomes.

# Backward induction

- ▶ Solve from the last decision nodes backward, folding in each node's optimal continuation value.
- ▶ In a finite game of perfect information this yields a subgame-perfect equilibrium.
- ▶ Backward induction is the engine behind Stackelberg, the entry game, and the chain store.

## UNIQUENESS CONDITION

The BI outcome is unique provided no player is ever indifferent — at one of their own decision nodes — between the folded-back values of two available actions. (Not "no terminal payoffs tie anywhere.")

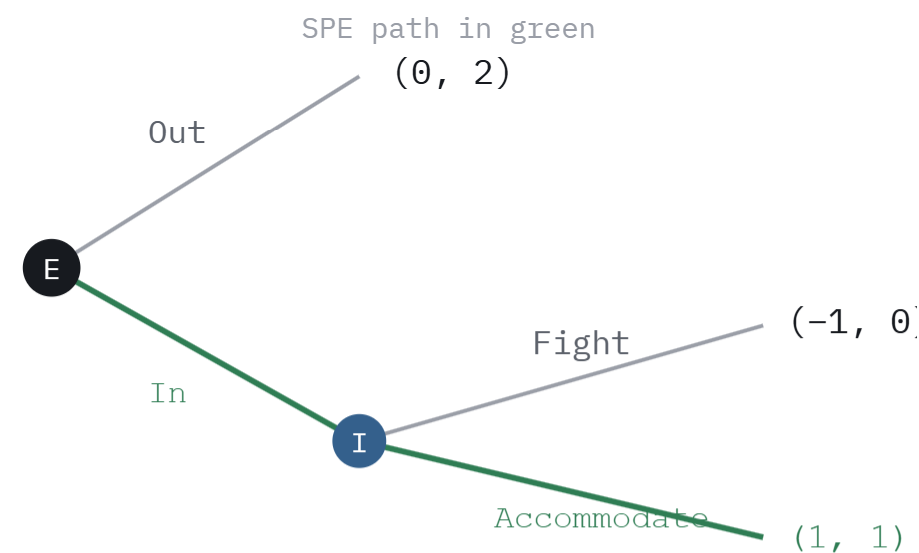
# Subgames & subgame-perfect equilibrium

- ▶ A subgame starts at a singleton information set and splits no information set across its boundary.
- ▶ The whole game is itself a subgame — which is exactly why every SPE is a Nash equilibrium.
- ▶ SPE rules out non-credible threats and promises.

## DEFINITION

$s^*$  is SPE  $\iff$  its restriction to every subgame is a Nash equilibrium.

# The entry game



|     | Fight | Accommodate |
|-----|-------|-------------|
| In  | -1, 0 | 1, 1 ★      |
| Out | 0, 2  | 0, 2        |

★ (In, Accommodate) is the unique SPE. (Out, Fight) is also Nash — a non-credible threat, since I's node is off-path and Nash checks optimality only on-path.

# Finitely repeated games unravel

- ▶ Total payoff  $U_i = \sum_{t=1}^T \delta^{t-1} g_i(a^t)$ .
- ▶ Theorem: if the stage game has a unique NE  $a^*$ , the unique SPE plays  $a^*$  every period, after every history.
- ▶ Multiple stage equilibria are necessary, not sufficient, for early non-NE play.
- ▶ The last period is always a stage NE (Benoît–Krishna) — full cooperation in period  $T$  is impossible.

## UNRAVELING

period  $T$  defects  $\Rightarrow$  period  $T-1$  defects  $\Rightarrow \dots \Rightarrow$  period 1 defects

# $T = 3$ defects; infinity need not

Stage game (row, col)

|   | C    | D    |
|---|------|------|
| C | 3, 3 | 0, 5 |
| D | 5, 0 | 1, 1 |

$D$  strictly dominates  $\Rightarrow$  unique stage NE  $(D, D)$ .

BACKWARD INDUCTION

Period 3  $\rightarrow (D, D)$ ; given that, period 2  $\rightarrow (D, D)$ ; then period 1  $\rightarrow (D, D)$   
. Total **3** each vs **9** under cooperation.

INFINITE HORIZON (grim trigger)

$$\frac{3}{1-\delta} \geq 5 + \frac{\delta}{1-\delta} \Rightarrow \delta \geq \frac{1}{2}$$

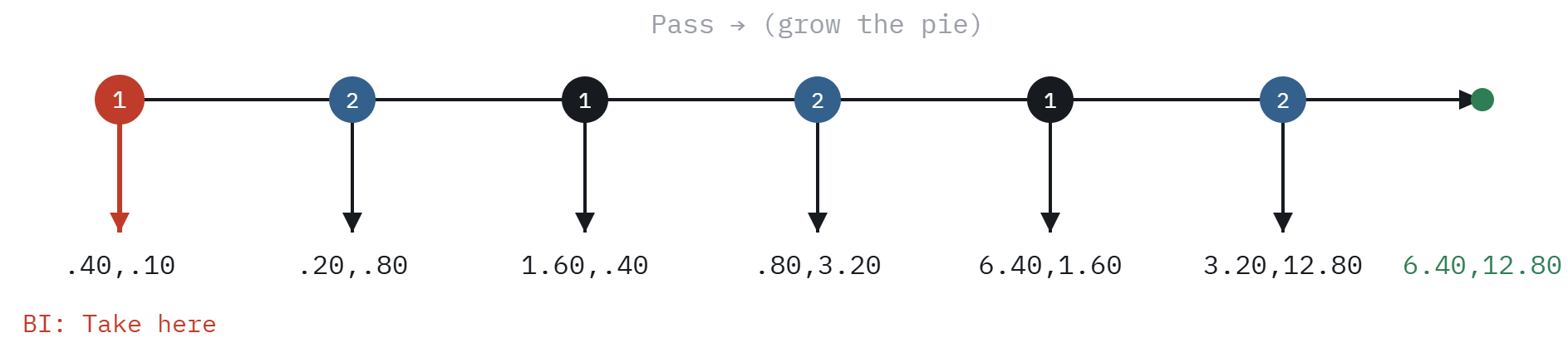
# The chain-store paradox

- ▶ A chain meets potential entrants in  $N$  towns, one after another.
- ▶ In the last town, fighting is unprofitable, so the incumbent accommodates.
- ▶ Backward induction unravels: accommodate everywhere, regardless of  $N$ .
- ▶ Resolution: incomplete information / reputation (Kreps–Milgrom–Roberts–Wilson).

## THE PUZZLE

Reputation seems valuable — yet with complete information and a known finite  $N$ , the threat to fight is never credible. A small chance the incumbent enjoys fighting restores deterrence.

# The centipede game



Folding back, every mover takes  $\Rightarrow$  **Take at node 1** gives (0.40, 0.10) — far below the cooperative (6.40, 12.80). Real players pass for several nodes (level- $k$  / QRE, social preferences, reputation).

# The one-shot-deviation principle

- ▶ A profile is SPE  $\iff$  no single-node deviation is profitable, holding the rest of the strategy fixed.
- ▶ Turns "check all deviations" into "check one-step deviations".
- ▶ Valid in finite games, and in discounted infinite games (continuity at infinity).
- ▶ It is exactly how we verify trigger strategies in Part D.

## IN WORDS

If you can't gain by deviating once and then reverting to the plan, you can't gain by any deviation at all.

# Centipede: when do you take?

Using the payoffs from the tree, predict how far play goes before someone takes.

What does theory say — and what will the room actually do?

► Reveal solution

# The pirate game

Five pirates, ranked by seniority, split 100 gold coins. The most senior proposes a division; all vote; if at least half approve it passes, otherwise the proposer walks the plank and the next-most-senior proposes. Pirates are greedy, rational, and bloodthirsty (prefer fewer rivals, all else equal).

How should the most senior pirate split the coins?

► Reveal solution

# Does a longer horizon help? $T = 10$ vs $T = 1,000,000$

Two firms play the prisoner's dilemma a known, finite number of times. Surely with a million rounds cooperation can get going?

Does the length of the horizon change the equilibrium?

► Reveal solution

PART D

# Infinitely Repeated Games & the Folk Theorem

Discounting, trigger strategies, the patience threshold  $\delta^*$  — and why the future is what makes collusion possible.

# Why infinity breaks unraveling

- ▶ No final period  $\Rightarrow$  backward induction has nowhere to start.
- ▶ After any history, the continuation game is identical to the whole game — stationarity.
- ▶ So future rewards and punishments carry real weight today.
- ▶ Cooperation becomes an equilibrium, not merely a hope.

## STATIONARITY

Every subgame after every history is isomorphic to the original game — the same threats are available forever.

# Discounting & the geometric sum

Payoff stream:  $U_i = \sum_{t=0}^{\infty} \delta^t u_i(a^t)$ ,  $0 < \delta < 1$ .

Let  $S = 1 + \delta + \delta^2 + \dots$ . Then  $\delta S = \delta + \delta^2 + \dots$ , so

$$S - \delta S = 1 \Rightarrow S = \frac{1}{1 - \delta}$$

Constant stream  $\pi \rightarrow \frac{\pi}{1 - \delta}$ ; the tail from period  $k$  is  $\frac{\delta^k \pi}{1 - \delta}$ .

## TWO READINGS OF $\Delta$

Time preference:  $\delta = \frac{1}{1 + r}$ .

Random termination: continue w.p.  $p$ .

If both apply, the effective factor is the product:

$$\delta = \frac{p}{1 + r}$$

# Trigger strategies

## Grim trigger

Cooperate until any defection, then play the stage NE forever (Nash reversion).

Credible because  $(D, D)$  is a stage NE — the punishment is itself an equilibrium.

## Tit-for-tat

Copy the opponent's last move — one-period memory, and forgiving.

Note: TFT is not generally subgame perfect off-path, so we verify with grim trigger.

# Sustaining cooperation in the PD

Stage payoffs:  $T > R > P > S$

Cooperate forever:  $\frac{R}{1-\delta}$ .

Best one-shot deviation:  $T + \frac{\delta P}{1-\delta}$ .

No-deviation condition  $\frac{R}{1-\delta} \geq T + \frac{\delta P}{1-\delta}$  rearranges to  $\delta(T - P) \geq T - R$ .

PATIENCE THRESHOLD

$$\delta \geq \delta^* = \frac{T - R}{T - P}$$

$0 < \delta^* < 1$  since  $R > P$

All-defect is an SPE for every  $\delta$ ; cooperation is the SPE needing  $\delta \geq \delta^*$ . The threshold is affine-invariant — only payoff-gap ratios matter.

# One-shot deviation & continuity at infinity

## ON-PATH

Deviating once gives  $T + \frac{\delta P}{1-\delta} \leq \frac{R}{1-\delta}$  exactly when  $\delta \geq \delta^*$ .

## OFF-PATH

After a defection both play  $D$  forever;  $(D, D)$  is a stage NE, so no one-shot deviation pays — the threat is credible.

## CONTINUITY AT INFINITY

Because stage payoffs are bounded and  $\delta < 1$ , far-future periods contribute negligibly — so checking one-shot deviations suffices for SPE.

# Feasible payoffs & the minmax

## FEASIBLE SET

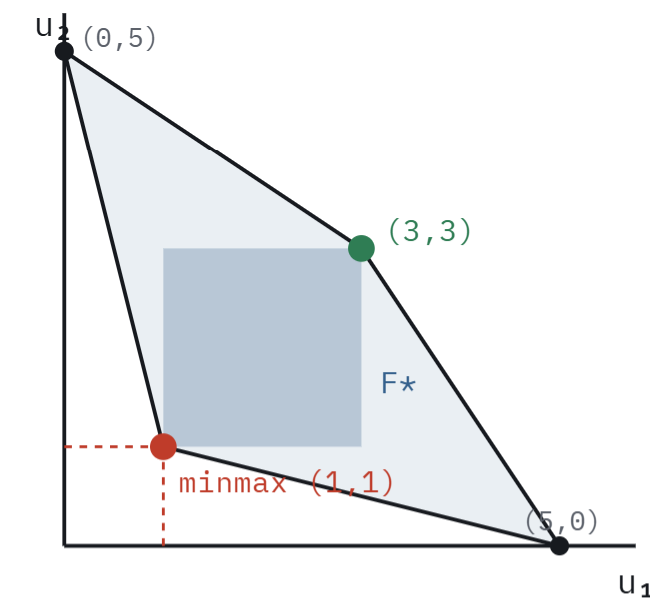
$F = \text{conv}\{u(a)\}$  – the convex hull of payoff vectors.

## MIXED-STRATEGY MINMAX

$\underline{v}_i = \min_{\alpha_{-i}} \max_{a_i} u_i(a_i, \alpha_{-i})$ .

## STRICTLY IR SET

$F^* = \{v \in F : v_i > \underline{v}_i \ \forall i\}$  – strictly above minmax.



# Two folk theorems

## Friedman (Nash reversion)

Any feasible payoff strictly above the static-Nash payoff is an SPE payoff for  $\delta$  near 1. Simple, weaker.

## Fudenberg–Maskin (minmax)

For any strictly-IR feasible  $v$ , if  $\dim F^* = N$  (full dimensionality), then  $\exists \bar{\delta} < 1$  s.t. for all  $\delta \geq \bar{\delta}$ ,  $v$  is an SPE payoff.

## INTUITION (un-normalized)

One-shot gain is fixed/finite; punishment loss  $\sim \frac{\delta}{1 - \delta} \rightarrow \infty$ . Patience makes deterrence overwhelming.

## LESSON & A NUANCE

Huge multiplicity: cooperation can be sustained, not that it will. Abreu's optimal penal codes punish toward minmax — harsher than Nash reversion — sustaining cooperation at lower  $\delta$ .

# The collusion threshold

GENERAL FORM

$$\delta^* = \frac{\pi^d - \pi^m / N}{\pi^d - \pi^N}$$

$\pi^d$  = best one-shot deviation against colluding rivals;  $\pi^m / N$  = collusive share;  $\pi^N$  = static-NE punishment.

Bertrand

$$\pi^d = \pi^m, \pi^N = 0 \Rightarrow \delta^* = \frac{1}{2}$$

Cournot

$$\delta^* = \frac{9}{17} \approx 0.53$$

Easier to collude with fewer firms, more frequent interaction, and faster detection.

# Two $\delta^*$ drills

- ▶ (i) PD with  $T = 10$ ,  $R = 6$ ,  $P = 2$ ,  $S = 0$
- ▶ (ii) Cournot collusion:  $P = 100 - Q$ ,  $c = 10$  , split monopoly

# Infinite PD: how patient must they be?

Stage payoffs  $T = 10$ ,  $R = 6$ ,  $P = 2$ ,  $S = 0$ . Under grim trigger, what is  $\delta^*$ ? And what if the punishment lasts only one period instead of forever?

Does a shorter punishment make cooperation easier or harder?

► Reveal solution

# Cournot collusion at $\delta = 0.5$ : sustainable?

Linear Cournot,  $P = 100 - Q$ ,  $c = 10$ , firms split the monopoly output and punish with Cournot reversion. At  $\delta = 0.5$ , can they hold the collusive price?

Compute  $\delta^*$  and compare it to 0.5.

► Reveal solution

# Summary

| Model                | Choice variable          | Price vs cost                   | Key condition / note                                                         |
|----------------------|--------------------------|---------------------------------|------------------------------------------------------------------------------|
| Monopoly / cartel    | quantity (joint)         | $P^m = \frac{a+c}{2}$ (highest) | not a NE — each firm wants to expand                                         |
| Cournot              | quantity (substitutes)   | $P^* = \frac{a+2c}{3}$ (middle) | $q^* = \frac{a-c}{3b}$ ; $\rightarrow$ competition as $N \rightarrow \infty$ |
| Bertrand (homog.)    | price                    | $P = c$ (lowest)                | paradox: 2 firms $\Rightarrow$ zero profit                                   |
| Bertrand (diff.)     | price (complements)      | $P = \frac{a+bc}{2b-d} > c$     | differentiation restores margin                                              |
| Stackelberg          | quantity, sequential     | below Cournot                   | leader commits, earns $2\times$ follower                                     |
| Repeated (collusion) | either, infinite horizon | up to $P^m$                     | sustainable iff $\delta \geq \delta^*$ (folk theorem)                        |

THE THROUGHLINE

# The future is the enforcement mechanism.

- ▶ In one shot, the strategic variable sets how far price sits above cost.
- ▶ A finite horizon unravels; an infinite or uncertain one revives cooperation.
- ▶ The folk theorem: patience makes cooperation possible — and predicts a multitude of outcomes.

● Monopoly   ● Cournot   ● Competition / Bertrand